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1 The Integers

Let $a - -b = c - -d$ if $a + d = b + c$. Equality is reflexive, symmetric, transitive. Define $(a - -b) + (c - -d) = (a + c) - -(b + d)$. $-$ is well-defined, commutative, associative. The set of $a - -b$ form an abelian group.

Define $(a - -b) \cdot (c - -d) = (ac + bd) - -(ac + bd)$. $\cdot$ is well-defined, commutative, associative (don't want to bash). $(\mathbb{Z}, +, \cdot)$ is a commutative ring with identity.

$\mathbb{Z}$ is an integral domain. Suppose $(a - -b) \cdot (c - -d) = 0 - -0$. Then $(ac + bd) - -(ad + bc) = 0 - -0$, so $ac + bd = ad + bc$. If $c \geq d$, let $c = d + \delta$. Then $a(d + \delta) + bd = ad + b(d + \delta) \rightarrow ad + a\delta + bd = ad + bd + b\delta \rightarrow a\delta = b\delta \rightarrow \delta = 0$ or $a = b$. If $\delta = 0$, then $c = d$. Else, $c \leq d$, so $d = c + \delta$. $ac + b(c + \delta) = a(c + \delta) + bc \rightarrow ac + bc + b\delta = ac + a\delta + bc \rightarrow b\delta = a\delta \rightarrow \delta = 0$ or $a = b$. If $\delta = 0$, $c = d$. Hence, $a - -b = 0 - -0$ or $c - -d = 0 - -0$.

We can embed $\mathbb{N}$ in $\mathbb{Z}$ by $a \mapsto a - -0$.

Define $a - -b \leq c - -d$ if $a + d \leq b + c$. $\leq$ is well-defined on $\mathbb{Z}$. $\leq$ is reflexive, transitive, antisymmetric, a total order (since $\mathbb{N}$ is a total order).

AM-GM over $\mathbb{N}$. $a^2 + b^2 \geq 2ab$ for all $a, b \in \mathbb{N}$. WLOG $a \geq b$, so $a = b + \delta$. Then $a^2 + b^2 = (b + \delta)^2 + b^2 = 2b^2 + 2b\delta + \delta^2 \geq 2(b + \delta)b = 2ab$.

AM-GM over $\mathbb{Z}$. $z^2 \geq 0$ for all $z \in \mathbb{Z}$. Let $z = a - -b$. Since $z^2 = (a^2 + b^2) - -(2ab)$, ISTS $a^2 + b^2 \geq 2ab$.

AM-GM over $\mathbb{Z}$. For all $z, w \in \mathbb{Z}$, we have $z^2 + w^2 \geq 2zw$. Factor as $(z - w)^2 \geq 0$.

General form: if $a - -b \geq 0$ and $x - -y \geq 0$, then $(a - -b) \cdot (x - -y) \geq 0$. Proof: let $a = b + d$ and $x = y + z$ and expand.

2 The Rationals

We define $\mathbb{Q} = \{a//b : (a, b) \in \mathbb{Z} \times \mathbb{Z}^\times\}$. Let $a//b = x//y$ if $ay = bx$, so equality is an ER. Define $m//n + x//y = (my + nx)/(ny)$. Define $m//n \cdot x//y = (mx)/(ny)$. Note if $n, y \neq 0$, then $ny \neq 0$ (since $\mathbb{Z}$ is an integral domain). $\mathbb{Q}$ has an identity, $+$ is commutative, associative, $\cdot$ distributes over $+$.

$\mathbb{Q}$ is a field.

For all rational $\epsilon > 0$, there exists rational $x > 0$ such that $x^2 < 2 < (x + \epsilon)^2$.

Consider the integer equation $p^2 - 2q^2 = 1$, for which $(3, 2)$ is a solution. Suppose (p_n, q_n) is a solution. I claim that $(p_{n+1}, q_{n+1}) = (3p_n + 4q_n, 2p_n + 3q_n)$ is a solution: $(3p_n + 4q_n)^2 - 2(2p_n + 3q_n)^2 = p_n^2 - 2q_n^2 = 1$. Further, $p_{n+1} \geq 3p_n$ and $q_{n+1} \geq 3q_n$, so $p_n, q_n \geq 3^{n-1}$.

If $p^2 - 2q^2 = 1$, then $\left(\frac{p}{q}\right)^2 = 2 + \frac{1}{q^2} > 2$.

If $\epsilon \geq 0.5$, then for $x = 1$, $x^2 < 2 < (x + 0.5)^2 \leq (x + \epsilon)^2$. Now assume $\epsilon < 0.5$. Note $p > q$. Consider $x = p/q - \epsilon > 0$. $x^2 = \left(\frac{p}{q} - \epsilon\right)^2 = \left(\frac{p}{q}\right)^2 - 2\epsilon\frac{p}{q} + \epsilon^2 = 2 + \frac{1}{q^2} - 2\epsilon\frac{p}{q} + \epsilon^2 < 2 + \frac{1}{q^2} - 2\epsilon + \epsilon^2 = 2 + \frac{1}{q^2} - \epsilon(2 - \epsilon)$. Pick q^2 such that $\frac{1}{q^2} < \epsilon(2 - \epsilon)$. Then $x^2 < 2 < (x + \epsilon)^2$, as desired.

Tao's proof: Sps FTSOC there exists $\epsilon > 0$ st $\forall x \geq 0$ we have not $x^2 < 2 < (x + \epsilon)^2$. In particular, if $x^2 < 2$, then $(x + \epsilon)^2 \leq 2 \rightarrow (x + \epsilon)^2 < 2$. Take $x = 0$, so $\epsilon^2 < 2$. Then $(\epsilon + \epsilon)^2 < 2$. By induction, $(n\epsilon)^2 < 2$. However, we may pick $\frac{2}{\epsilon} < n$, so $n\epsilon > 2 \rightarrow (n\epsilon)^2 > 4 > 2$, cont.

3 Cauchy Sequences

A sequence (x_n) is said to be Cauchy if for all $\epsilon > 0$ there exists N st $i, j \geq N \rightarrow |x_i - x_j| < \epsilon$. A sequence (x_n) is said to converge to $q \in \mathbb{Q}$ if for all $\epsilon > 0$, there exists N st $n \geq N \rightarrow |x_n - q| < \epsilon$.

Every convergent sequence is Cauchy. Let $\epsilon > 0$. Pick N st $n \geq N \rightarrow |x_n - q| < \epsilon/2$. Let $i, j \geq N$. Then $|x_i - x_j| \leq |x_i - q| + |x_j - q| < \epsilon$, so (x_n) is Cauchy.

Every subsequence (x_{a_n}) of a Cauchy sequence (x_n) is Cauchy. Let $\epsilon > 0$. Pick N st $i, j \geq N \rightarrow |x_i - x_j| < \epsilon$. Then since $a_i \geq i \geq N$ and $a_j \geq N$, $|x_{a_i} - x_{a_j}| < \epsilon$.

Every subsequence (x_{a_n}) of a convergent sequence (x_n) is convergent. Let $\epsilon > 0$.

Pick N st $n \geq N \rightarrow |x_n - q| < \epsilon$. Since $a_n \geq n \geq N$, we have $|x_{a_n} - q| < \epsilon$.

Every Cauchy sequence (x_n) is bounded. Pick N st $i, j \geq N \rightarrow |x_i - x_j| < 1$. Then $x_i \in (x_N - 1, x_N + 1)$ for all $i \geq N$. Pick M st $x_i \in (-M, M)$ for $i \in [N - 1]$. Pick K st $\{x_i\} \subset (-K, K)$. Corollary: every convergent sequence is bounded.

If $(x_n), (y_n)$ are Cauchy, so is $(x_n + y_n)$. Let $\epsilon > 0$. Pick N st $i, j \geq N \rightarrow |x_i - x_j| < \epsilon/2, |y_i - y_j| < \epsilon/2$. Then $|x_i + y_i - x_j - y_j| \leq |x_i - x_j| + |y_i - y_j| < \epsilon$.

If $(x_n), (y_n)$ are Cauchy, so is $(x_n y_n)$. $(x_n), (y_n)$ are both bounded, say $\{x_n\}, \{y_n\} \subset (-M, M)$. Let $0 < \epsilon < 1$. Pick N st $i, j \geq N \rightarrow |x_i - x_j| < \frac{\epsilon}{2M+1}, |y_i - y_j| < \frac{\epsilon}{2M+1}$. Then

$$\begin{aligned} |x_i y_i - x_j y_j| &\leq |x_i| |x_i - x_j| + |y_i| |y_i - y_j| + |x_i - x_j| |y_i - y_j| \\ &\leq M \frac{\epsilon}{2M+1} + M \frac{\epsilon}{2M+1} + \epsilon \\ &= \frac{2M+1}{2M+1} \epsilon = \epsilon. \end{aligned}$$

If $(x_n - x'_n)$ converges to 0 and (y_n) is bounded, then $((x_n - x'_n)y_n)$ converges to 0. Pick M st $\{y_n\} \subset (-M, M)$. Let $\epsilon > 0$. Let N st $n \geq N \rightarrow |x_n - x'_n| < \epsilon/M$. Then $|(x_n - x'_n)y_n| = |x_n - x'_n| |y_n| < \frac{\epsilon}{M} M = \epsilon$.

4 The Reals

Let $\mathbb{R}$ denote the set of all Cauchy $\mathbb{Q}$ -sequences. Give $\mathbb{R}$ the ER that $x = y$ if $(x_n - y_n)$ converges to 0.

$=$ is an ER. $x = x$. If $x = y$, then $(x_n - y_n) \rightarrow 0$, so $(y_n - x_n) \rightarrow 0$, so $y = x$. If $x = y$ and $y = z$, then $(x_n - y_n) \rightarrow 0$ and $(y_n - z_n) \rightarrow 0$, so $(x_n - z_n) = (x_n - y_n + y_n - z_n) \rightarrow 0 + 0$.

If $x, y \in \mathbb{R}$, let $x + y = (x_n + y_n)$. Since $(x_n), (y_n)$ are Cauchy, so is $(x_n + y_n)$. If $x' = x$, then $(x'_n - x_n) \rightarrow 0$, so $(x'_n + y_n - x_n - y_n) \rightarrow 0$, so $x' + y = x + y$.

If $x, y \in \mathbb{R}$, let $x \cdot y = (x_n y_n)$. Since $(x_n), (y_n)$ are Cauchy, so is $(x_n y_n)$. If $x = x'$, then $(x'_n - x_n) \rightarrow 0$. Since (y_n) is Cauchy, (y_n) is bounded, so $((x'_n - x_n)y_n) \rightarrow 0$, i.e. $(x'_n y_n - x_n y_n) \rightarrow 0$, or $x' \cdot y = x \cdot y$.

Let $x \in \mathbb{R}$. We say $x > 0$ if there exists $\epsilon > 0$ st there exists N fw $n \geq N \rightarrow x_n \geq \epsilon$. Define $x < 0$ similarly.

Trichotomy of $\mathbb{R}$. Let $x \in \mathbb{R}$. Then $x < 0, x = 0$, or $x > 0$. Suppose $x \neq 0$. Then

there exists $\epsilon > 0$ such that there doesn't exist N st $n \geq N \rightarrow |x_n| < \epsilon$. Let $N_S = \{n \in N : x_n \in S\}$. Consider the partition of N : $N_{(-\infty, \epsilon]}, N_{(-\epsilon, \epsilon)}, N_{[\epsilon, \infty)}$. If either $N_{(-\infty, \epsilon]}, N_{[\epsilon, \infty)}$ are finite, let N be larger than the greatest element of $N_{(-\infty, \epsilon]} \cup N_{[\epsilon, \infty)}$. Then $n \geq N \rightarrow x_n \in N_{(-\epsilon, \epsilon)}$, or $|x_n| < \epsilon$, cont. Hence, at least one of $B = N_{(-\infty, -\epsilon]}, T = N_{[\epsilon, \infty)}$ are infinite. WLOG T is infinite.

Sps FTSOC $L = N_{(-\infty, \epsilon/2)}$ is infinite. Pick $a_1 \in T$. Pick $a_2 \in L$ st $a_2 > a_1$ (since L is infinite). Define a_n recursively, alternating between T, L . Since (x_n) is Cauchy, so is (x_{a_n}) . Pick N st $i, j \geq N \rightarrow |x_{a_i} - x_{a_j}| < \epsilon/2$, cont as $x_{a_N}, x_{a_{N+1}}$ are in different sets of T, M and differ by at more than $\epsilon/2$. Hence, L is finite. Let N be larger than the max of L . Then $n \geq N \rightarrow n \in N_{[\epsilon/2, \infty)}$. That is, $x > 0$. The cases $x > 0, x = 0, x < 0$ are disjoint.

If $S \subset N$ and S is infinite, if $x \in \mathbb{Q}$, there exists $s \in S$ st $x < s$.

Pick $n \in N$ st $x < n$. Define (a_n) recursively by $a_n = \min(S - a([n-1]))$. Since $a([n-1])$ is finite, $a([n-1]) \subsetneq S$. N is well-ordered, so a_n is well-defined. By construction, $a_n \neq a_1, a_2, \dots, a_{n-1}$, so $a : N \rightarrow N$ is injective. Sps FTSOC for $n < m$ that $a_n > a_m$. Note $a_m \notin \{a_1, a_2, \dots, a_{m-1}\}$, so $a_m \notin \{a_1, a_2, \dots, a_{n-1}\}$, cont min of a_n . Hence $a : N \rightarrow N$ is strictly increasing, so $a_n \geq n$. Take the same n as before. Then $x < n \leq a_n$, for $a_n \in S$.

$+$ is commutative. $x + y = (x_n + y_n) = (y_n + x_n) = y + x$. Similarly, $\cdot$ is commutative. $+$ is associative. $(x + y) + z = (x_n + y_n) + z = ((x_n + y_n) + z_n) = (x_n + (y_n + z_n)) = x + (y_n + z_n) = x + (y + z)$. Similarly, $\cdot$ is associative. $x \cdot (y + z) = x \cdot (y_n + z_n) = (x_n \cdot (y_n + z_n)) = (x_n \cdot y_n + x_n \cdot z_n) = (x_n \cdot y_n) + (x_n \cdot z_n) = x \cdot y + x \cdot z$. Hence, $\cdot$ distributes over $+$.

$x + 0 = x$. $x + (-x) = 0$. $x \cdot 1 = x$. Hence, $(\mathbb{R}, +, \cdot)$ is a commutative ring with identity.

If $x \neq 0$, there exists $y \in \mathbb{R}$ st $xy = 1$. WLOG $x > 0$, so there exists κ, M st $n \geq M \rightarrow x_n \geq \kappa$. Let y be a sequence st $y_n = 1/x_n$ for $n \geq M$.

Claim: (y_n) is Cauchy. Let $\epsilon > 0$. Pick $N \geq M$ st $i, j \geq N \rightarrow |x_i - x_j| < \kappa^2 \epsilon$. Then $|\frac{1}{x_i} - \frac{1}{x_j}| = \frac{|x_i - x_j|}{|x_i||x_j|} < \frac{\kappa^2 \epsilon}{\kappa^2} = \epsilon$, so (y_n) is Cauchy. Further, $xy = 1$. $\mathbb{R}$ is a field.

Note if $x > 0, y > 0$, then $x + y > 0$. Say $x \geq y$ if $x - y \geq 0$. I claim $\geq$ is a total order.

$\geq$ is reflexive. $x \geq x$ since $x - x = 0 \geq 0$. $\geq$ is antisymmetric: sps $x \geq y$ and $y \geq x$, so $x - y \geq 0$ and $y - x \geq 0$, so $x - y \leq 0$. Then $(x - y = 0$ or $x - y > 0)$ and $(x - y = 0$ or $x - y < 0)$. By Trichotomy of $\mathbb{R}$, $x = y$.

$\geq$ is transitive. If $x \geq y$ and $y \geq z$, then $x - y \geq 0$ and $y - z \geq 0$. If $x - y = 0$ and $y - z = 0$, then $x - z = (x - y) + (y - z) = 0$. If $x - y > 0$ and $y - z = 0$, then $x - z = (x - y) + (y - z) = x - y > 0$. If $x - y = 0$ and $y - z > 0$, then $x - z = (x - y) + (y - z) = y - z > 0$. If $x - y > 0$ and $y - z > 0$, then $x - z = (x - y) + (y - z) > 0$.

If $x, y > 0$, then $x \cdot y > 0$.

If $x > y$ and $z > 0$, is $xz > yz$? $x - y > 0$, so $(x - y) \cdot z > 0$, so $x \cdot z - y \cdot z > 0$, so $x \cdot z > y \cdot z$.

If $x > y$, $z < 0$, then $xz < yz$. $x(-z) > y(-z)$, so $x(-z) - y(-z) > 0$, so $yz - xz > 0$, so $yz > xz$.

Note that $\mathbb{Q} \hookrightarrow \mathbb{R}$ since $q = (q, q, q, \dots)$. If $x < y$, then $x \cdot 2 < y + x$, so $x < (x + y)/2$. Similarly, $(x + y)/2 < y$, so $\mathbb{R}$ is a linear continuum.

Between any two distinct reals there lies a rational. Sps $x \neq y \in \mathbb{R}$. WLOG $x > y$, so there exists $M, \epsilon > 0$ for which $n \geq M \rightarrow x_n - y_n \geq \epsilon$. Take $N \geq M$ such that $i, j \geq N \rightarrow |x_i - x_j| < \epsilon/3, |y_i - y_j| < \epsilon/3$. Let $q = \frac{x_N + y_N}{2}$. Note in particular, $x_N - y_N \geq \epsilon > 0$. Let $U = (x_N - \epsilon/3, x_N + \epsilon/3)$, $V = (y_N - \epsilon/3, y_N + \epsilon/3)$. Note $\{x_n\}_{n \geq N} \subset U, \{y_n\}_{n \geq N} \subset V$ and U, V are disjoint.

Consider the rational $p = (q, q, q, \dots)$. $x > p$: if $n \geq N$, then $x_n - q \geq x_N - \epsilon/3 - q \geq x_N - (x_N - y_N)/3 - (x_N + y_N)/2 = (x_N - y_N)/6$. Similarly, $p > y$.

If $x \in \mathbb{R}$, there exists $q > x$ with $q \in \mathbb{Q}$. Pick $\{x_n\} \subset (-M, M)$. Then $q = (M + 1, M + 1, \dots) > x$.

Let $S \neq \emptyset$ be bounded above. Pick $x_0, y_0 \in \mathbb{Q}$ such that $[x_0, \infty) \cap S \neq \emptyset$ and y_0 is an upper bound of S . If $x_0 = y_0$, we're done. Else, define intervals $I_0 \supset I_1 \supset I_2 \supset \dots$ as follows. Let $I_0 = [x_0, y_0]$. At least one of the intervals $[x_0, (x_0 + y_0)/2], [(x_0 + y_0)/2, y_0]$ intersects S . If it's the second, take $x_1 = (x_0 + y_0)/2$ and $y_1 = y_0$. Otherwise, take $x_1 = x_0$ and $y_1 = (x_0 + y_0)/2$. In general, $y_n - x_n = (y_0 - x_0)/2^n$, y_n is an upper bound for S , and I_n intersects S . Note further than (x_n) is nondecreasing, (y_n) is nonincreasing, $x_n \leq y_n$ for all n .

I claim $(x_n), (y_n) \in \mathbb{R}$. Let $\epsilon > 0$. Pick $N > (y_0 - x_0)/\epsilon$. Then for $i, j \geq N$, $|x_i - x_j| \leq (y_0 - x_0)/2^N \leq (y_0 - x_0)/N < \epsilon$, so (x_n) is Cauchy. The case is similar for y . Further, (for similar reasons), $(x_n - y_n) \rightarrow 0$, so $x = y$.

Sps FTSOC y is not an ub of S , so there exists $s \in S$ st $y < s$. Let $\epsilon > 0, N$ be st $n \geq N \rightarrow s_n - y_n \geq \epsilon$. Pick $M \geq N$ st $\{y_n\}_{n \geq M} \subset (y_M - \epsilon/2, y_M + \epsilon/2)$. Since y_M is an ub for S , $y_M \geq s$. If $y_M = s$, we're done. Else, pick $\epsilon' > 0, L \geq M$ st $n \geq L \rightarrow y_M - s_n \geq \epsilon'$. In particular, $y_M - s_L \geq \epsilon'$. $y_L \in (y_M - \epsilon/2, y_M + \epsilon/2)$,

and $s_L - y_L \geq \epsilon$. The first and last inequalities give $y_M - y_L \geq \epsilon + \epsilon'$. The second implies $y_M - y_L < \epsilon/2$, cont.

Sps FTSOC there exists $b' < b$ st b' is an ub for S . Pick N st $(y_N - x_N) \leq (y_0 - x_0)/2^N < b - b'$. Pick $s \in S$ such that $s \in [x_N, y_N]$. Then $b' < (b - y_N) + x_N \leq x_N \leq s$, so $b' < s$, i.e., b' is not an ub of S , cont. Hence b is the sup of S .

Hole: if $b = (b_n) \in \mathbb{R}$ st $q_n = (b_n, b_n, b_n, \dots)$ is ub for S , then b is ub for S .